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Quantifying the carbon footprint reduction potential of lifestyle choices in Japan

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E-mail: keiichiro.kanemoto@gmail.com**Keywords:** household carbon footprint, lifestyle choices, Japan, MRIO, regression analysisSupplementary material for this article is available [online](#)

Abstract

Numerous studies have investigated the hotspots for reducing carbon emissions associated with household consumption, including reducing household carbon footprints (CFs) and greener lifestyle choices, such as living car-free, eating less meat, and having one less child. However, estimating the effect of each of these actions requires the simultaneous consideration of lifestyle choices and household characteristics that could also affect the household CF. Here, we quantify the reduction in household CFs for 25 factors associated with individual lifestyle choices or socioeconomic characteristics. This study linked approximately 42 000 microdata on consumption expenditure with the Japanese subnational 47 prefecture-level multi-regional input–output table, which are both the finest-scale data currently available. We improved the accuracy of household CF calculations by considering regional heterogeneity, and successfully estimated the magnitude of household CF reduction associated with individual lifestyle choices and socioeconomics. For example, it was found that moving from a cold region to a region with mild climate would have considerable potential for reducing the CO₂ emissions of a household, all other factors being equal. In addition, a household residing in a house that meets the most recent energy standards emits 1150 kg less CO₂ per year than if they reside in a house that meets previous energy standards. Ownership and use of durable goods also had the potential for reducing the CO₂ emissions of a household; a normal-sized car, a personal computer, a compact car, and a bidet were associated with CO₂ emissions of 922, 712, 421, and 345 kg per year, respectively. The findings therefore have important implications for climate change mitigation and policy measures associated with lifestyle.

1. Introduction

The consumption of goods generates direct and indirect (i.e. lifecycle) environmental and resource impacts in their supply chains. In recent decades, the impacts of goods and services have been widely assessed in consumption-based accounting as ‘footprints’ (Wiedmann and Lenzen 2018, Heinonen *et al* 2020). Studies of environmentally extended input–output analysis (IOA) have shown that daily household consumption is the dominant contributor to greenhouse gas (GHG) footprints (i.e. carbon footprint (CF)), accounting for around two-thirds of global carbon emissions (Hertwich 2011,

Ivanova *et al* 2016). Numerous studies have therefore investigated the important consumption drivers of the household CF, and examined what personal actions could reduce this footprint effectively, such as living car-free, eating less meat, and having fewer children (Wynes and Nicholas 2017, Vita *et al* 2019, Ivanova *et al* 2020). Some of these studies focused on household characteristics to identify which attributes of a household contribute most to household CFs, and most established that higher-income households have larger footprints than lower income households (Wiedenhofer *et al* 2018). These studies mainly examined the effect of each household characteristic on the household CF.

It is also important to consider the specific consumption choices in conjunction with the household attributes being controlled for. For instance, high-income households in a city would increase their CF by using more electric appliances, and decrease their footprint by avoiding car travel because of the well-developed public transport infrastructure in cities.

Previous studies have examined the relationship between household income and carbon and energy footprints using econometric approaches (e.g. Lenzen *et al* 2006, Baiocchi *et al* 2010, Wiedenhofer *et al* 2013, Jiang *et al* 2020). While these analyses provided insights into reduction policies related to CFs, the sample sizes used were small because only the average expenditure in each area was utilized, which meant that the spatial resolution was insufficient for accurately characterizing factors that impact the household CF (Ivanova *et al* 2017). The use of aggregated (e.g. country- or regional-level) data is problematic because it can lead to misspecification and/or biased estimates (Theil 1954, Orcutt *et al* 1968, Clark and Avery 1976). To deal with these problems of data aggregation, some studies used datasets containing data from a large number (e.g. more than 1000) of households (Ala-Mantila *et al* 2014, 2016, Baiocchi *et al* 2015, Ottelin *et al* 2015, 2019, Fremstad *et al* 2018, Gill and Moeller 2018, Koide *et al* 2019, Li *et al* 2019), or large datasets were constructed for analysis (Jones and Kammen 2014). For example, Fremstad *et al* (2018) utilized a quarterly panel dataset with approximately 28 000 households in the US and found that CFs could potentially be reduced through an economy of scale, which provides opportunities for carbon-intensive goods to be shared. Jones and Kammen (2014) applied econometric models to approximately 30 000 household consumption expenditures in national household survey data linked to US postal codes. They also clarified the relationships between household CFs and US socioeconomic and typological factors, such as population density, the number of rooms in a house, and the age of a house, all of which can influence consumption. However, due to the national input–output model adopted, the CF estimates obtained in these studies did not reflect differences between subnational regions with regard to the technologies and supply chains that produced the consumption goods and services.

In this study, we examined the relationship between household characteristics (e.g. socioeconomic, geographic, and demographic measures) and household CFs in Japan using a rich dataset. The dataset was constructed by combining a subnational multi-regional input–output model (MRIO) with micro-survey data, which included 25 characteristics for approximately 42 000 households. To our knowledge, this is the largest and most accurate dataset currently available in Japan. We also set a goal of

quantifying the CF reduction for each lifestyle choice. The findings of this study can, therefore, provide more detailed insights into the factors that potentially affect the household CF, with useful policy implications for GHG emissions reductions associated with changes in lifestyles as demand-side solutions (Seto *et al* 2016, Creutzig *et al* 2018).

2. Methods

2.1. Data for detailed estimations of household CF

The household CF is derived from direct carbon emissions from fossil fuel combustion (i.e. driving a car) and indirect carbon emissions associated with household electricity use and the production of goods and services consumed through household supply chains (i.e. passenger car manufacturing) (Weber and Matthews 2008). In a review, Tukker *et al* (2010) examined income level, household size (number of family members per household), geographic location, house type, automobile ownership, food consumption patterns, international (and interregional) trade, and social and cultural differences as the key determinants of the household CF. In addition, they found that for economically developed nations, the most critical consumption categories affecting the household CF were food and beverages, mobility, housing, and products that use energy, such as household appliances. This result has been corroborated by other studies (Hertwich 2005, 2011, Huppes 2006, Tukker and Jansen 2006, Ivanova *et al* 2016, 2017, Shigetomi *et al* 2017). Based on these studies, we quantified the spatial attributes of household CFs in Japan in detail, and conducted a regression analysis of household energy and CFs based on previous studies.

The variables used for the regression analysis and their definitions and sources are listed in table 1. The household consumption expenditures used to estimate the CF and most of the explanatory variables were retrieved from the microdata of the Japanese National Survey of Family Income and Expenditure (NSFIE) for the year 2004, with special permission from the Ministry of Internal Affairs and Communications, Japan (MIC). The dataset consists of information on 60 058 households. The population density, *Density*, was determined for each city where the household is located. The numerator of *Density*, i.e. total population, was obtained from the 2004 Basic Resident Registration of Japan (MIC 2011); in this way the population data corresponded to the microdata of the NSFIE. The denominator, i.e. area of each region, was calculated using geographical information system data provided by the National Land Numerical Information, and downloaded from the Ministry of Land, Infrastructure, Transport and Tourism (MLIT 2020). House type was classified as an apartment or freestanding house; the dummy variable value for *House* is 0 (apartment) or 1 (freestanding house). Since the Japan Meteorological Agency (2017) does

Table 1. Descriptive statistics: cross-sectional data for 2004.

Variable	Unit	Obs.	Mean	Median	Std. Dev.	Min.	Max.	Explanation	Reference
Explained									
<i>CF_all</i>	t/household	60 058	14.99	13.39	8.274	0.787	155.4	Annual CF per household for all goods and services	Estimated based on the raw data of NSFIE (MIC 2017) with the 47 prefecture MRIO (Hasegawa <i>et al</i> 2015) and energy balance table (METI 2018) in this study.
<i>CF_food</i>	t/household	60 058	2.938	2.759	1.341	0.0312	24.89	Annual CF per household for food	
<i>CF_electricity</i>	t/household	60 058	0.667	0.595	0.372	0	4.477	Annual CF per household for electricity	
<i>CF_gas</i>	t/household	60 058	1.127	1.026	0.667	0	10.97	Annual CF per household for gas	
<i>CF_other heating</i>	t/household	60 058	0.363	0.000	0.836	0	21.74	Annual CF per household for kerosene and heat	
<i>CF_durable goods</i>	t/household	60 058	1.782	1.171	2.427	0	101.6	Annual CF per household for household durable goods	
<i>CF_consumable goods</i>	t/household	60 058	1.469	7.364	1.835	0	114.8	Annual CF per household for household consumable goods	
<i>CF_education</i>	t/household	60 058	0.736	0.000	2.960	0	137.1	Annual CF per household for education	
<i>CF_medicals</i>	t/household	60 058	0.410	0.256	0.622	0	26.06	Annual CF per household for medical products and services	
<i>CF_private transport</i>	t/household	60 058	2.136	1.601	2.244	0	44.77	Annual CF per household for passenger vehicles including motor bikes and passenger cars	
<i>CF_public transport</i>	t/household	60 058	0.753	0.137	1.562	0	43.54	Annual CF per household for public transport	
<i>CF_other services</i>	t/household	60 058	2.605	2.070	2.325	0	56.63	Annual CF per household for information, recreation, rent, and other services	
Explanatory									
<i>Income</i>	10 ⁴ JPY	60 058	608.7	527.0	470.1	0	12 643	Annual household income	NSFIE (MIC 2017)
<i>Savings</i>	10 ⁴ JPY	60 058	1382	760.0	1922	0	50 770	Household savings	NSFIE (MIC 2017)
<i>Density</i>	People/km ²	59 882	2916	930.9	5088	2.746	50 209	Population density where the household is located at the city level	National Land Numerical Information (MLIT 2020)
<i>CDD</i>	C° · day	60 058	6.169	4.876	7.032	0	91.64	Monthly CDDs	Japan Meteorological Agency (2017)
<i>HDD</i>	C° · day	60 058	18.44	4.493	29.86	0	260.7	Monthly HDDs	Japan Meteorological Agency (2017)
<i>Floor</i>	m ²	60 058	122.8	111.5	78.70	7	3300	Gross floor areas in house	NSFIE (MIC 2017)
<i>Built I</i>	(Dummy)	42 382	0.320	1	0.467	0	1	1: when the owned house was built during 1980–1993; 0: otherwise	NSFIE (MIC 2017)
<i>Built II</i>	(Dummy)	42 382	0.133	0	0.339	0	1	1: when the owned house was built during 1994–1997; 0: otherwise	NSFIE (MIC 2017)

(Continued.)

Table 1. (Continued.)

Variable	Unit	Obs.	Mean	Median	Std. Dev.	Min.	Max.	Explanation	Reference
<i>Built III</i>	(Dummy)	42 382	0.165	0	0.371	0	1	1: when the owned house was built in 1998 or later; 0: otherwise	NSFIE (MIC 2017)
<i>House</i>	(Dummy)	60 058	0.757	0	0.429	0	1	1: freestanding house; 0: apartment	NSFIE (MIC 2017)
<i>Child</i>	Person	60 058	0.673	0	0.993	0	6	Number of children aged 0–18 years in household	NSFIE (MIC 2017)
<i>EmployAdult</i>	Person	60 058	1.079	1	0.888	0	6	Number of employed individuals aged 18–64 years in household	NSFIE (MIC 2017)
<i>UnemployAdult</i>	Person	60 058	0.747	1	0.726	0	6	Number of unemployed individuals aged 18–64 years in household	NSFIE (MIC 2017)
<i>Employ65</i>	Person	60 058	0.108	0	0.371	0	3	Number of employed individuals aged more than 65 years in household	NSFIE (MIC 2017)
<i>Unemploy65</i>	Person	60 058	0.478	0	0.729	0	4	Number of unemployed individuals aged more than 65 years in household	NSFIE (MIC 2017)
<i>VehicleNS</i>	Unit	60 058	1.029	1	0.804	0	5	Number of normal sized vehicles (>660 cc)	NSFIE (MIC 2017)
<i>VehicleK</i>	Unit	60 058	0.411	0	0.656	0	5	Number of light vehicles ($\leq$ 660 cc) that a household owns	NSFIE (MIC 2017)
<i>Motorbike</i>	Unit	60 058	0.192	0	0.477	0	5	Number of motor bikes that a household owns	NSFIE (MIC 2017)
<i>AirCon</i>	Unit	60 058	2.218	2	1.739	0	9	Number of air conditioners that a household owns	NSFIE (MIC 2017)
<i>Microwave</i>	Unit	60 058	1.009	1	0.349	0	6	Number of microwaves that a household owns	NSFIE (MIC 2017)
<i>PC</i>	Unit	60 058	0.922	1	0.927	0	9	Number of personal computers that a household owns	NSFIE (MIC 2017)
<i>Fridge</i>	Unit	60 058	1.254	1	0.613	0	8	Number of refrigerators that a household owns	NSFIE (MIC 2017)
<i>TV</i>	Unit	60 058	2.180	2	1.308	0	12	Number of televisions that a household owns	NSFIE (MIC 2017)
<i>Washer</i>	Unit	60 058	1.059	1	0.378	0	5	Number of washing machines that a household owns	NSFIE (MIC 2017)
<i>Bidet</i>	Unit	60 058	0.684	1	0.705	0	7	Number of bidets that a household owns	NSFIE (MIC 2017)

not report city-level temperatures, we estimated the heating degree days (HDDs) and cooling degree days (CDDs) for each prefecture as the simple average of values observed at all the meteorological weather stations within a prefecture.

The NSFIE database lists the year that the house was built (e.g. 2000). In Japan, the law of the standard for home energy saving was passed in 1980, and the heat insulation requirements for housing were amended; for example, energy conservation standards were updated in 1994, 1998, 2013, and 2016. Thus, depending on the year when the house was built, we used three dummy variables to distinguish between the home energy standards. *Built I*, *Built II*, and *Built III* take on a value of 1 if the house was built during 1980–1993, 1994–1997, and 1998 or later, respectively. The base group of these dummy variables consists of houses built during 1955–1979. Due to the survey design of the NSFIE, the year of construction for a household is listed only if the household owned their house and it was built in 1955 or later. Consequently, data were missing for 17 676 observations, and the number of observations used in our main model was therefore 42 257. In order to use as many observations as possible for each regression equation, those households with missing values were omitted from the analysis, which meant that the number of observations ranged from 42 257 to 60 058 (see tables S4 and S6 (available online at stacks.iop.org/ERL/16/064022/mmedia)). We also analyzed the regression equations by using the smallest ‘common’ dataset to examine whether omitting households with missing values influenced our regression results (see tables S8 and S9 in the supporting information (SI)).

To obtain the explained variables, we used an MRIO model covering the 47 prefectures in Japan (47MRIO) (Hasegawa *et al* 2015) to estimate the CO₂ emissions embodied per unit of expenditure (CF intensity). The MRIO model is a recent solution for distinguishing the regional differences in technology and supply chain structures. MRIO tables describe economic transactions across multiple regions, and have been adopted to quantify the CF, as well as carbon leakage from rapid development in the recent decade (Wiedmann *et al* 2011, Inomata and Owen 2014, Lenzen *et al* 2017, Naeyegele and Zaklan 2019). Most recently, Ivanova *et al* (2017) and Ottelin *et al* (2019) used an MRIO model to conduct a regression analysis for estimating household CFs across nations in the European Union (EU). The 47MRIO comprises 80 commodities and 47 prefectures based on 2005 data. Although the targeted year is vintage, the model describes the commodity sectors in more detail than other interregional models (e.g. Chinese MRIO of Mi *et al* 2018 has 30 economic sectors for 30 provinces). Our analysis is the first to combine subnational MRIO and micro-consumption data to analyze the CF of households in Japan.

2.2. Estimating household CFs using a regression model

The household CF, Q_{ij} , due to consumption of commodity j by household i , was quantified based on the IOA (i.e. Leontief demand-pull model; Miller and Blair 2009) using equation (1):

$$Q_{ij} = e_{pj}f_{ij}, \quad (1)$$

where, e_{pj} refers to the total (i.e. direct and indirect) CO₂ emissions per unit consumption expenditure on commodity j in prefecture p where household i lives, and f_{ij} to the consumption expenditure on commodity j by household i . The data for f_{ij} were obtained from the NSFIE, while those for e_{pj} were estimated based on the 47MRIO and sectoral direct carbon intensities, which we will explain below.

The carbon intensity, e_{pj} , can be decomposed into two terms, $e_{pj} = e_{pj}^d + e_{pj}^i$, where e_{pj}^d and e_{pj}^i are direct and indirect emissions, respectively. To quantify the direct intensity, e_{pj}^d , we considered only energy-related commodities (i.e. gasoline, light oil, kerosene, liquefied petroleum gas, city gas, and coal products). The national average direct carbon intensity (Nansai and Moriguchi 2012) was applied due to the limited data availability. For indirect emissions, e_{pj}^i , we calculated, for the first time, the 2005 carbon intensities for the 47 prefectures in Japan using the 47MRIO, the domestic energy balance table by prefecture, and the results of a direct survey of each prefectural government. After determining the CF intensity for each prefecture and each commodity, we matched 80 commodities that determine the carbon intensity to 320 items on the NSFIE (table S3). Then, consumption expenditures were transformed from the purchaser price to the producer price to make them consistent with the carbon intensity using the national margin table. Given that the NSFIE is published every five years, we used the NSFIE from 2004, the closest publication year to the 47MRIO, to obtain the specific consumption expenditures by household. The method used to calculate the household CF used in this study is also elaborated (Kanemoto *et al* 2019, 2020).

Next, to examine the relationships between the CF and the household characteristics selected for this study, we used the regression equation formulated in equation (2):

$$\ln(Q_{Ci}) = \beta_0 + \mathbf{x}_i\boldsymbol{\beta} + \epsilon_i, \quad (2)$$

where $Q_{Ci} = \sum_{j \in C} Q_{ij}$ denotes the CF of household i . C indicates a consumption category or a group of commodities. We adopted the following categories and considered the differences among them in terms of the drivers of CFs: (a) food and beverages, (b) electricity, (c) gas, (d) other heating (e.g. kerosene), (e) durable goods, (f) consumable goods, (g) education

(e.g. electricity related to school activities), (h) medical (e.g. operation of medical equipment), (i) private transport, (j) public transport, and (k) other services. In this study, these 11 categories ($C = 1, \dots, 11$) were determined by aggregating 320 commodity sectors ($j = 1, \dots, 320$) listed in the Japanese NSFIE (MIC 2017) in line with the ‘classifications by goods and services’ defined in the NSFIE (see table S3). Durable goods cover home electric appliances, furniture, recreational equipment, bicycles, and bags. Note that passenger vehicles and their associated products and fuels are classified as private transport. Consumable goods include nondurable items, except food and beverages, medicines and supplements, and fuels. Other services include non-goods not attributed to (a)–(j) (e.g. commercial laundry, water supply and sewerage, information and communication services). $\mathbf{x}_i = (x_{ik})$ is the vector for the explanatory variables for the attributes of household i . β_0 and β are parameters to be estimated, and ϵ_i is an error term. The explanatory variables selected in this study, and the related hypotheses based on the previous studies, are presented by domain in table 2 and in the SI (see sections S1.2 and 1.3). We transformed the values for income, savings, population density, and gross floor area as well as CFs into their logarithmic forms. Note that a value of 1 was added to income, savings, and CF before taking the logarithm because those variables are 0 for some households. This would change the basis of the rate of change from 1 to 0 (Wooldridge 2020).

3. Results and discussion

3.1. Regression analysis of the Japanese household CF

The ordinary least squares (OLS) results obtained by regression analysis are shown in table 3. Additionally, to better clarify the relationship between the CF and household characteristics, we applied the OLS regression method to the household CFs for the 11 consumption categories. The estimation results are presented in table S6. We also applied the seemingly unrelated regression (SUR) for the purposes of comparison; we selected the SUR to consider possible correlations between the error terms in the 11 regression equations (see table S7). For both tables S6 and S7, selected explanatory variables were excluded from the regression equations if they were regarded as irrelevant. For instance, the variables indicating the ownership of vehicles were included only in the equations for private and public services; all of the variables are included in the equation for other services.

Overall, most of the coefficients in the main model shown in table 3 were statistically significant. We examined the robustness of the results obtained using the main model in table 3, by excluding those variables that were not significant (i.e. *Motorbike*,

Microwave, *Fridge*, and *Washer*). The estimated coefficients for the significant variables were not substantially changed and they remained significant, as shown in table S5. We also calculated the variance inflation factor (VIF) and confirmed that there was no serious multicollinearity problem (i.e. VIFs were ≤ 2.30). R-squared was approximately 0.385, which is comparable with the adjusted R-squared reported in a previous study (Koide *et al* 2019), which analyzed Japanese CFs using microdata and the global MRIO for Japan. Here, we summarize our most important findings.

The coefficients for $\ln(\text{Income})$ and $\ln(\text{Income})^2$ were both significant and positive in the main model, suggesting that the CF would increase progressively with income growth. Note that the correlations for the squared values of *Income* and *Savings* were used in the regression equations to examine whether the CF increases nonlinearly with income and might decrease after a certain income threshold (Baiochi *et al* 2010, Ivanova *et al* 2017). Table 3 shows that the CF increased monotonically with *Income*. Regarding the elasticity of the CF with respect to income, the estimates obtained using the sample mean were significantly positive among all of the columns (See table S2). Moreover, *Savings* showed a similar trend to *Income*, which is associated with an increase in the CF.

Next, the coefficient for $\ln(\text{Density})$ was significantly negative, which is consistent with the previous studies listed in table 2. An increase in population density, which is a proxy for urbanization, is estimated to generally reduce the household CF. Conversely, looking at the results by consumption category (table S6), population density was associated with an increase in the CFs for food, education, and public transport, which indicates the specific impacts of urbanization on the household CF and identifies targets for mitigation.

The coefficient for *Built III* was also significantly negative, indicating that newer (i.e. more energy efficient) houses are associated with a reduction in the CF. As expected, the coefficient for $\ln(\text{Floor})$ was significantly positive, which is consistent with previous studies (e.g. Lenzen *et al* 2006). The finding that the coefficient for *House* was significantly negative was surprising as this implies that living in a freestanding house is superior to living in an apartment in terms of the CF. This is a unique, or probably unrealistic, result because a freestanding house should not be superior to an apartment in terms of energy efficiency, particularly in terms of air conditioning (see table 2). Estimating the CF by consumption category (table S6) showed that the coefficient for *House* was significantly positive for other types of heating (e.g. kerosene). This result is consistent with our assumption that a freestanding house is less efficient than an apartment for air conditioning. Conversely, the coefficient for *House* was significantly negative for

Table 2. Selected explanatory variables and expected coefficients used for estimating the household CF.

Domains	Indicator	Predicted effect	Reasoning	References
Economics	<i>Income</i>	+	Household income is understood to be a positive determinant of household consumption expenditure because people can spend more on goods and services with a higher income.	(Lenzen et al 2006, Baiocchi et al 2010, Tukker et al 2010, Minx et al 2013, Ala-Mantila et al 2014, Ala-Mantila et al 2016, Jones and Kammen 2014, Ivanova et al 2017, Ivanova et al 2018, Fremsted et al 2018, Gill and Moeller 2018, Li et al 2019, Koide et al 2019)
	<i>Savings</i>	+	The amount of savings indicates the degree of allowance for consumption. Higher savings are associated with more purchases of expensive durable goods and services such as leisure travel.	Koide et al (2019)
Urbanization	<i>Density</i>	±	Population density is a useful proxy for degree of urbanization. Energy requirements for private transport and residence would be reduced while that for public transport increases with higher population density. However, urban areas have higher carbon impacts from food, leisure, and manufactured products.	(Lenzen et al 2006, Wiedenhofer et al 2013, Minx et al 2013, Ala-Mantila et al 2014, Ala-Mantila et al 2016, Jones and Kammen 2014, Ottelin et al 2015, Ottelin et al 2018b, Ottelin et al 2019, Gudipudi et al 2016, Ivanova et al 2017, Chen et al 2018, Gill and Moeller 2018, Li et al 2019, Koide et al 2019)
Physical characteristics of dwellings	<i>House (dummy)</i>	+	An apartment has better insulation than a solitary house due to fewer outside walls and, generally, smaller rooms, as well as more energy efficient building standards.	(Lenzen et al 2006, Tukker et al 2010, Ivanova et al 2018, Koide et al 2019)
	<i>Built I (dummy)</i>	±	Newer houses built with higher energy standards for building have better insulation, which is expected to reduce electricity, gas, and heating use.	(Jones and Kammen 2014, Ottelin et al 2015)
	<i>Built II (dummy)</i>			
	<i>Built III (dummy)</i>			
	<i>Floor</i>	+	Larger floor areas or more rooms would require more home energy use and appliances.	(Lenzen et al 2006, Baiocchi et al 2010, Jones and Kammen 2014, Fremstad et al 2018, Ivanova et al 2018, Li et al 2019)
Local climate	<i>CDD</i>	+	As the monthly CDDs increase, home energy use from gas stoves and other heaters is likely to be higher.	(Mansur et al 2008, Zhou and Gurney 2011)
	<i>HDD</i>	+	As the monthly HDDs increase, home energy use from air conditioning is likely to be higher.	(Mansur et al 2008, Zhou and Gurney 2011, Wiedenhofer et al 2013, Ivanova et al 2017)

(Continued.)

Table 2. (Continued.)

Domains	Indicator	Predicted effect	Reasoning	References
Demographics	<i>Child</i>	+	A larger family size increases household consumption. In particular, more children lead to greater expenditures related to education. The number of unemployed people in a household is expected to raise home energy use because the daytime hours that the house is occupied are longer. Medical expenses are likely to increase in households with people aged more than 65 years.	(Lenzen <i>et al</i> 2006, Mansur <i>et al</i> 2008, Baiocchi <i>et al</i> 2010, Tukker <i>et al</i> 2010, Ala-Mantila <i>et al</i> 2014, Ala-Mantila <i>et al</i> 2016; Ottelin <i>et al</i> 2015, Ottelin <i>et al</i> 2018b, Ottelin <i>et al</i> 2019, Ivanova <i>et al</i> 2018, Huang <i>et al</i> 2019, Koide <i>et al</i> 2019)
	<i>EmployAdult</i>			
	<i>UnemployAdult</i>			
	<i>Employ65</i>			
	<i>Unemploy65</i>			
Ownership of private vehicles	<i>VehicleNS</i>	±	The number of private vehicles changes the CF associated with fuel and maintenance. Also, ownership of private vehicles reduces the frequency of public transport use.	(Tukker <i>et al</i> 2010, Minx <i>et al</i> 2013, Wiedenhofer <i>et al</i> 2013, Gill and Moeller 2018, Li <i>et al</i> 2019, Koide <i>et al</i> 2019)
	<i>VehicleK</i>			
	<i>Motorbike</i>			
Ownership of home appliances and electronics	<i>AirCon</i>	+	It is expected that the greater the number of electric appliances and home electronics owned, the more electricity is used. The number of home appliances and electronics would increase with household income.	Koide <i>et al</i> (2019)
	<i>Microwave</i>			
	<i>PC</i>			
	<i>Fridge</i>			
	<i>TV</i>			
	<i>Washer</i>			
	<i>Bidet</i>			

Table 3. Regression results for total household CF using OLS method.

Domains	Variables	
Economics	$\ln(\text{Income})$	−0.299 ^{***} (0.006)
	$\ln(\text{Income})^2$	0.0478 ^{***} (0.0008)
	$\ln(\text{Savings})$	−0.0306 ^{***} (0.0034)
	$\ln(\text{Savings})^2$	0.00525 ^{***} (0.00036)
Urbanization	$\ln(\text{Density})$	−0.00632 ^{***} (0.00146)
Dwellings	<i>House</i>	−0.0821 ^{***} (0.00651)
	<i>Built I</i>	0.00763 [*] (0.0045)
	<i>Built II</i>	−0.0355 ^{***} (0.0059)
	<i>Built III</i>	−0.0767 ^{***} (0.0057)
	$\ln(\text{Floor})$	0.0394 ^{***} (0.0056)
Local climate	<i>CDD</i>	0.000693 ^{**} (0.000345)
	<i>HDD</i>	0.000810 ^{***} (0.000080)
Demographics	<i>Child</i>	0.0332 ^{***} (0.0019)
	<i>EmployAdult</i>	0.0101 ^{***} (0.0031)
	<i>UnemployAdult</i>	0.0701 ^{***} (0.0030)
	<i>Employ65</i>	−0.0514 ^{***} (0.0059)
	<i>Unemploy65</i>	0.0178 ^{***} (0.0032)
Ownership of private vehicles	<i>VehicleNS</i>	0.0615 ^{***} (0.0030)
	<i>VehicleK</i>	0.0281 ^{***} (0.0031)
	<i>Motorbike</i>	0.00545 [*] (0.00363)
Ownership of home appliances and electronics	<i>AirCon</i>	0.0176 ^{***} (0.0014)
	<i>Microwave</i>	0.00738 [*] (0.00631)
	<i>PC</i>	0.0475 ^{***} (0.0022)
	<i>Fridge</i>	0.00549 [*] (0.00352)
	<i>TV</i>	0.00803 ^{***} (0.00169)
	<i>Washer</i>	−0.0204 ^{***} (0.0059)
	<i>Bidet</i>	0.0230 ^{***} (0.0029)
	<i>Constant</i>	2.253 ^{***} (0.031)
	Observations	42 257

Table 3. (Continued.)

R-squared	0.385
BIC	33 641

Standard errors calculated by the Huber-White method are in parenthesis.

The adjusted R-squared is equal to the R-squared, at least, up to the third decimal place.

BIC is the Schwarz Bayesian information criterion.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

electricity. This does not necessarily imply that our assumption is incorrect, because electricity is used for a variety of purposes and not limited to air conditioning, but we were unable to clarify the reasons for this result. It is therefore necessary to explore other factors that affect the coefficient for *House* that reduce the total household CF, such as the building structure and other home equipment, which were not included in this study.

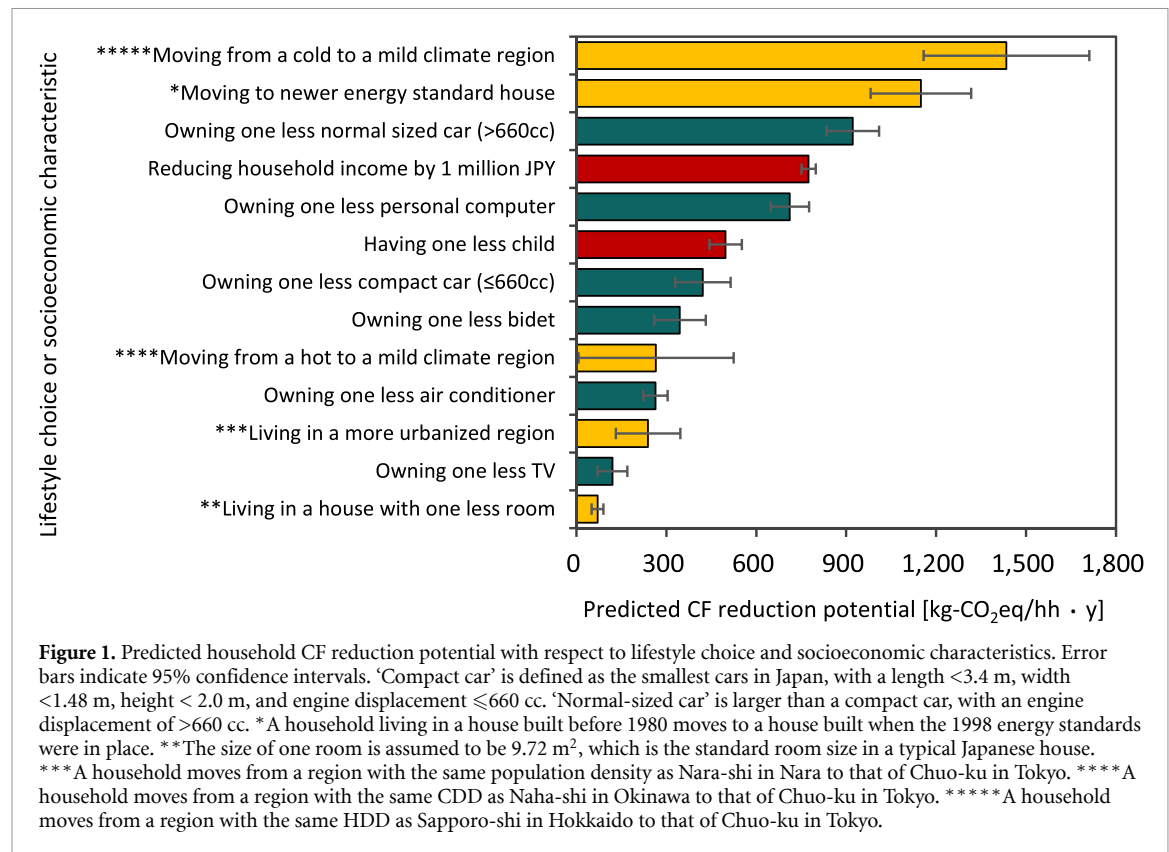
Regarding the demographic factors, *Child*, *EmployAdult*, *UnemployAdult*, and *Unemploy65*, were all positively correlated with the household CF, which is expected. The coefficient for *UnemployAdult* was associated with the largest increase in the household CF and was followed by *Child*, *Unemploy65*, and *EmployAdult*. On the other hand, the coefficient for *Employ65* was negative. Taken together, these differences are expected to reflect the length of time spent at home. More detailed analyses are elaborated in sections S2.1 and S2.2.

3.2. Effective lifestyle choices for carbon reduction

To determine which factors should be prioritized to mitigate climate change in line with lifestyle choice, we estimated the potential CF reduction achieved under typical lifestyle choice scenarios based on the regression results discussed above. For example, the scenario ‘living in a more urbanized region’ is associated with a change in population density between two regions with other factors being equal. Details of the analysis method are given in section S1.4 in the SI.

Figure 1 summarizes the potential household CF reductions by lifestyle choice or socioeconomic characteristics that were calculated based on the estimation results obtained using the main model shown in table 3.

Note that each of these potential reductions indicate how lifestyle choice affects the decrease in the household CF, under the assumption that all other factors are kept constant. The potential reductions vary across households because our regression is nonlinear. We therefore estimated the potential



reductions using the sample mean (see section S1.4). These lifestyle choices were selected based on those examined in previous studies (Wynes and Nicholas 2017, Koide *et al* 2019, Ivanova *et al* 2020) and the statistical significance in this study; for example, ‘having one less child’ was identified as having the highest impact on carbon reduction by Wynes and Nicholas (2017). We do not recommend implementing these actions without careful consideration of how these actions will affect the other factors than CF.

In figure 1, our estimates indicate that moving from a cold region to a region with a mild climate has the greatest effect on carbon reduction, with a difference of as much as 1435 kg-CO₂/household-year (hh-yr). Followed by this, the potential footprint reduction that can be achieved by living in a house with the newer energy standards also presented the large potential reduction (figure 1). If a household moves from a house built before 1980 (i.e. a house constructed with no energy standards) to a house built after 1998 (i.e. a house with the most recent energy standards), then the potential footprint reduction would be 1150 kg-CO₂/hh-yr, which represents the second largest CF reduction potential. On the other hand, moving from a hot region to a mild region has a much lesser effect on carbon reduction than moving from a cold region to a mild region (265 kg-CO₂/hh-yr). Note that in estimating the reduction potential of local climate shown in figure 1, regions with cold, hot, and mild climates are those that have CDD and HDD values equal to those

in Okinawa (the southernmost prefecture), Hokkaido (the northernmost prefecture), and Tokyo (the capital, located around the center of Japan). These three estimations indicate the importance of energy saving for heating a house.

The third largest potential footprint reduction (922 kg-CO₂/hh-yr) was achieved by owning one less normal-sized car. Owning one less compact car also reduces the CF, but not by as much as one less normal-sized car (421 kg-CO₂/hh-yr). Therefore, reducing the dependence on private transport is essential for reducing the CF. Policy changes to support carbon mitigation by decreasing car ownership could include improvements in public transport infrastructure and implementing carpooling incentives. In addition, encouraging the purchase of smaller vehicles by increasing the tax on larger vehicles, such as the car weight tax implemented in Japan since 1971, would also be effective for reducing the household CF.

Ownership of electric appliances/home electronics also had a positive impact on the CF (figure 1). In particular, owning one less personal computer is expected to reduce the CF by as much as the reduction potential that can be achieved by decreasing the annual household income by 1 million JPY (≈9042 US dollars) (712 and 775 kg-CO₂/hh-yr, respectively). Interestingly, owning one less bidet would reduce the CF of a household by more than one air-conditioner and a TV, perhaps because a bidet is likely to be kept on all the time. It is recommended that people decrease the frequency of use of these appliances by

not leaving PCs in sleep mode for extended periods or using bidets only during the winter.

In terms of demographic factors, this study supports the results of previous studies which showed that reducing the number of children in a household is a positive driver for the household CF. As shown in figure 1, the CF reduction potential of having one less child is 498 kg-CO₂/hh-yr. Note that this reduction potential would be offset by a number of potentially simpler lifestyle choices, including owning one less car or owning one less personal computer.

Finally, living in a more urbanized region is defined as a move from a region with the same population density as Nara (a commuter city in a suburban region) to Tokyo (the capital and most populous city in Japan) under the assumption that all of the regional characteristics, except population density, are the same. Such a choice is equivalent to a reduction potential of 239 kg-CO₂/hh-yr. Related to the changes associated with living in a house, a decrease in the number of rooms does not appear to affect CF reduction (we here assume that the floor area of one room is 9.72 m², which is the standard room size in Japan) (figure 1). Note that although moving to a freestanding house from an apartment is associated with a CF reduction, as shown in table 3, we did not estimate this reduction potential and present it in figure 1 because this finding warrants further clarification; see also the results for the robustness checks in section S2.2.

3.3. Limitations

There are several limitations associated with calculating household CFs by combining the IOA with the survey-based NSFIE data. First, we assumed that the CF of one Japanese yen's worth of an imported commodity was equivalent to that of the corresponding domestic commodity. Second, the quality of commodities consumed by households could not be considered. For instance, distinguishing between domestic and international air travel would affect the household CF significantly (Czepkiewicz *et al* 2018). Third, government expenditure for public welfare services, such as health care and education, and capital formation were not included in the estimates, resulting in the underestimation of CFs induced by actual consumption (Heinonen *et al* 2020). These are general limitations associated with IOA approaches, and some could be addressed if the relevant data were available (Ottelin *et al* 2018a, Berrill *et al* 2020, Schmidt *et al* 2019); however, it is currently impossible to obtain some of these data for each region. Another limitation related to data availability is that the NSFIE and the 47MRIO datasets are not perfectly matched with each other with respect to geographical resolution (e.g. city

versus prefecture) and time scales (e.g. 2004 versus 2005). Further, because the data used in our analysis are from 2005, this study does not consider up-to-date technology, such as solar panels, electric vehicles, and smart phones, that might affect current household energy and CFs. These limitations are associated with MRIO data availability. Future research efforts will focus on addressing the aforementioned limitations and expanding our analysis to factors that drive household CFs in other nations and globally.

4. Conclusion

This study analyzed the drivers of the Japanese household CF with respect to socioeconomics, demographics, urbanization, physical dwelling characteristics, local climate, ownership of passenger vehicles, home appliances and electronics, by combining microdata for more than 42 000 households' consumption and an MRIO model for Japan. We successfully estimated the magnitude of the household CF reduction associated with individual lifestyle choices and socioeconomic factors. We further provided new insights into the lifestyle choices that have the greatest potential for reducing a household's CF by using CFs disaggregated into multiple consumption categories. Identifying these carbon-reducing lifestyle choices has important policy implications. For example, the findings related to demographic factors are relevant to both climate change policy and aging society policy in Japan (Shigetomi *et al* 2014, Prime Minister of Japan and His Cabinet 2016, Shigetomi *et al* 2018). More broadly, our analysis provides quantitative information on potential CF reductions as they relate to policy measures. While having one less child is known to reduce the household CF, we show that other, more feasible, lifestyle choices, such as owning one less car, could more effectively reduce the CF. Policies could be established to support these lifestyle choices by means of subsidies, taxation, and infrastructure improvements.

Data availability statement

The data generated and/or analysed during the current study are not publicly available for legal/ethical reasons.

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